A Unified Physics-Aware Quantum Machine Learning Framework across Power GaN HEMTs and Logic Nanowire FETs: Predicting Unseen Process Splits and Held-Out Geometry Combinations with Lower Error and Tighter Split-to-Split Variability

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Abstract- We present a unified reinforcement-learning (RL) framework that discovers compact parametrized quantum circuits (PQCs) for data-scarce device modeling. A graph neural network (GNN) policy optimized by proximal policy optimization (PPO) searches circuit architectures using leave-one-group-out cross-validation (LOGOCV) error on held-out process or geometry groups as the reward. The framework achieves the lowest mean absolute error (MAE) on all 11 targets versus six classical baselines, with 59% lower error (I_{off}) and 81% tighter fold variability (V_{TH}) for HEMTs and 84% lower error (V_{TH}, SS, I_{off}) and 82% tighter fold variability (I_{off}) for NWFETs. These results demonstrate the potential of RL-selected, classically simulated PQCs as compact surrogates with low OOD error and improved physical consistency, despite imposing no explicit physical constraints, penalty terms, or device-specific equations, on the two evaluated device datasets.

I. INTRODUCTION

Advanced device architectures are evolving to meet the distinct demands of high-voltage power conversion and sub-3-nm logic. p-GaN gate GaN HEMTs enable normally-off operation with controllable threshold voltage for high-performance power switching [1], while gate-all-around nanowire FETs (NWFETs) suppress short-channel effects and off-state leakage for continued scaling toward low-power logic [2].

Their optimization is limited by costly fabrication and TCAD cycles and by stochastic variation from charge trapping, thermal gradients, and parameter extraction. High-value design exploration thus requires physically consistent extrapolation to unfabricated process splits and held-out geometry combinations, not merely interpolation among measured devices (**Fig. 1**).

Existing surrogates often embed device-specific equations in the architecture [16]-[18], impeding reuse across device families, or report only within-distribution tests [16][17]. We instead impose no explicit physical constraints, penalty terms, or device-specific equations, making out-of-distribution (OOD) prediction the explicit objective; physical consistency emerges from the OOD reward. Parametrized quantum circuits (PQCs) provide a compact Fourier-structured function class whose expressivity grows with depth and data re-uploading [7][8] rather than parameter count [9]. A relational graph-attention policy represents candidate circuits as heterogeneous graphs, optimized by PPO with a leave-one-group-out cross-validation reward, selecting architectures for unseen process or geometry groups rather than average training fit.

A single search framework applies to both planar HEMTs and 3D NWFETs without re-deriving device equations, discovering a circuit optimized for each (**Fig. 4**): HEMTs use physics-based pretraining followed by calibration to measured wafer data, while NWFETs learn directly from 3D TCAD. Across 11 targets, the selected circuits reduce OOD error by 6-34% for HEMTs and 25-39% for NWFETs, with about 5× tighter HEMT split-to-split variability. To our knowledge, this is the first experimentally validated, RL architecture-searched PQC framework [10] for multi-target device modeling optimized directly for OOD prediction (**Table 3**).

II. METHODOLOGY

Dataset preparation: To simulate p-GaN HEMTs for pretraining, a 1D electrostatic solver incorporating Mg-acceptor ionization, polarization charge, 2D electron-gas (2DEG) formation, and interface traps [3]–[5] (**Fig. 5**) was developed. Solved by finite-volume discretization with damped Newton iteration, it extracts full transfer curves and six targets: forward/reverse threshold voltages ($V_{TH,for}$/$V_{TH,rev}$), hysteresis (ΔV_{TH}), subthreshold swing (ss), on-current (I_{ON}), and off-current (I_{OFF}). Electrostatic analysis confirms physical validity (**Fig. 3**): smooth band bending, gate-modulated surface potential, enhancement-mode operation with expected 2DEG densities, and charge conservation at the sweep turnaround. The solver generates 1,785 devices/h on a CPU (~90× faster than TCAD), while a PhysicsNeMo PINN reaches 12,898 devices/h on a GPU (~650× faster) [6] (**Table 3**) to augment it. The experimental HEMT dataset comprises 544 devices spanning unique process splits and wafer positions; the NWFET dataset comprises 1,914 3D-TCAD devices for five targets across eight groups defined by channel length, nanowire radius, and dielectric thickness.

Encoding and circuit search: Device features map one-to-one to qubits by angle encoding: five process/position inputs for HEMTs and three geometry inputs (channel length (L_g), radius (r), dielectric thickness (t)) for NWFETs. The agent selects a pre-encoder, R_X, R_Y, or R_Z rotations, and entanglers from CNOT, CRX, CRY, and CRZ families arranged as chain, ring, all-to-all, or even/odd pairs. The shared library contains 5 rotation and 24 entangling blocks plus data re-uploading, with candidate depth 3-12 blocks and at most four re-uploads. Validity masks require at least one rotation and one entangler, forbid consecutive re-uploads, and allow early termination, limiting unproductive exploration.

Readout: Single-qubit Z expectations pass through a trainable tempered tanh and linear map to standardized multi-target outputs. For measured HEMTs, a wafer-position correction is added to the quantum prediction, letting the PQC capture process-dependent behavior while the small residual head accounts for spatial drift across the wafer.

GNN policy and RL: Each partial circuit is encoded as a typed graph with feature, qubit, operation-block, and readout nodes linked by circuit-specific relations (**Fig. 2**). A relational graph attention network serves as a shared actor-critic backbone [11], pooling masked node embeddings with global circuit features. The actor selects the pre-encoder, feature encoding, next valid block, or termination; the critic estimates terminal reward. PPO with generalized advantage estimation updates the policy [12][13], and a 0.001 per-block penalty discourages unnecessary depth. This preserves connectivity across variable-length candidates and device-specific qubit counts.

Training and selection. HEMT training uses three stages: synthetic electrostatic pretraining, wafer-position-head calibration with the PQC frozen, and joint experimental fine-tuning; NWFETs use direct single-stage TCAD training. Architecture selection follows nested LOGOCV: the inner reward uses the worst validation-group error with complexity and invalid-structure penalties, so a strong average cannot hide failure on one split; the selected candidate is evaluated on an untouched outer group. Hybrid quantum-classical training minimizes mean-squared error alone, with no physics-informed loss or constraint terms, using parameter-shift gradients [14] on NVIDIA CUDA-Q [15].

III. RESULTS AND DISCUSSION

We compare our RL-selected hybrid quantum neural network (PQC) against six classical baselines on two OOD challenges: measured p-GaN HEMT process variation and 3D NWFET geometry scaling (**Fig. 7**). All models use the same group-wise validation protocol to prevent data leakage; results are averaged over 5 random seeds.

To isolate the model-class effect, the ANN baseline used the identical data pipeline, wafer-position correction, training stages, and LOGOCV splits as the PQC. It was swept across shallow-to-deep capacities plus a Fourier-feature ablation, and we report its best variant.

Benchmark accuracy and efficiency: The PQC has the lowest OOD MAE on all 11 targets versus the six classical baselines. For HEMTs, $V_{TH,for}$ MAE is 18.6% below the identically trained ANN and 48.0% below the worst baseline (PLS); $V_{TH,rev}$ MAE is 25.2% below the ANN and 42.1% below the worst baseline; and I_{OFF} MAE is 23.5% below the ANN and 59% below the worst baseline (Elastic Net). ΔV_{TH}, SS, and I_{ON} MAE also improve, while forward threshold-voltage fold variability is 53% tighter than the ANN and 81% tighter (5.3×) than the worst baseline (Decision Tree). For NWFETs, the PQC leads on all five targets, V_{TH}, SS, and I_{OFF} MAE are 21.4%, 29.7%, and 37.5% below the ANN, respectively, and about 84% below the worst baseline (PLS); I_{OFF} fold variability is 82% tighter than the worst baseline (**Fig. 13)**. It uses 2,460 trainable model parameters (4.7× fewer than the ANN) and sits on the best error-parameter Pareto frontier (**Fig. 8**).

Physical consistency in HEMTs: Although PQC and ANN have similar aggregate error distributions, the PQC better preserves threshold-hysteresis closure (**Fig. 9**), even though no hysteresis-closure constraint or physics-based penalty is applied during training, so this consistency arises from the OOD-rewarded architecture search rather than explicit enforcement. Consistency metrics pool out-of-fold predictions from models refitted per outer group. We measure the absolute deviation of ($V_{th,rev}$ - $V_{th,forw}$) from the independently predicted hysteresis (ΔV_{TH}), which should be zero for consistent outputs. The PQC median consistency error is about 5.4× lower than the ANN's; the ANN's inconsistent tail reaches roughly 4× its own median (about 22× the PQC median). Residual decomposition shows recipe-level mean bias falls to about 0.12 interquartile range (IQR) versus about 0.4 IQR for ANN, whereas die-to-die pattern error stays similar at about 0.4-0.5 IQR. The improvement mainly reflects suppressed wafer-to-wafer drift without losing local variation.

Retention of pretrained physics after fine-tuning: To test whether calibration preserves the process behavior acquired during solver pretraining, the fine-tuned process cores were reevaluated on the synthetic corpus with the wafer-position residual head excluded (**Fig. 11**). Before fine-tuning, both models retained all 16 evaluable solver-trend signs, with mean Spearman agreement ρ_s = 0.949 (ANN) and 0.935 (PQC). After fine-tuning, ANN sign retention fell to 75% and mean agreement to ρ_s = 0.501, reversing all four I_{on} trends, whereas PQC retained all 16 signs with mean ρ_s = 0.875. PQC also exceeded ANN in sign retention and trend agreement across all LOGO folds, with substantially lower fine-tuning-induced degradation.

Joint-geometry OOD generalization in NWFETs: To evaluate electrostatic gate control across subthreshold and on-state regimes, we model three metrics: off-current, subthreshold swing, and maximum transconductance. For these radius-sensitive targets, the PQC reduces out-of-fold MAE relative to the ANN by 50.1%, 46.3%, and 39.2%, respectively. Because weak-inversion current is exponentially sensitive to small potential-barrier errors in held-out geometries, we bin normalized MAE by distance to the nearest training geometry (**Fig. 10**). Both models degrade deeper into OOD space, but the PQC degrades more slowly: its advantage grows from 45.5% (I_{OFF}) and 30.0% (SS) in the nearest bin to 51.6% and 58.9% in the deepest bin. The widening margin indicates the quantum prior captures geometry-dependent electrostatic scaling rather than memorizing sampled points. To verify that OOD gains reflect genuine physical consistency, we evaluate preservation of geometry-dependent TCAD behavior (**Fig. 12**). Along controlled slices with two geometry variables fixed, both models reproduce the dominant trends, but only the PQC tracks highly nonlinear responses such as the non-monotonic g_{max} dependence on dielectric thickness (t). Across all exact slices, the PQC achieves higher directional agreement with TCAD than the ANN (0.928 vs. 0.903) and a finite-difference slope ratio much closer to unity (0.979 vs. 0.877). The PQC also yields lower normalized MAE in 30 of 32 target–geometry combinations, with a median relative reduction of 42.2% and a worst-octant value of 0.133 vs. 0.333. Because these folds withhold joint low/high combinations of L_g, r, and t while using individual axis values seen elsewhere in training, the framework generalizes robustly to unseen joint geometries and missing design-space corners.

IV. CONCLUSION

We demonstrate the first experimentally validated, RL-searched PQC framework optimized directly for OOD device prediction. **Table 4** highlights that, for the first time, the proposed framework uniquely combines multi-target OOD prediction, RL-searched quantum circuits, and scalability across distinct power and logic device families without embedding device-specific equations**.** A single physics-aware workflow generalizes across measured p-GaN HEMT process splits and unseen 3D-NWFET geometry combinations, achieving the lowest OOD MAE for all 11 targets among six classical baselines. Compared with the identically trained ANN, the PQC reduces MAE by up to 50.1% while using 4.7× fewer parameters; versus all baselines, reductions reach 59% for HEMTs and 84% for NWFETs. Crucially, it preserves all 16 pretrained HEMT physical trends, improves threshold–hysteresis consistency by 5.4×, and maintains nonlinear NWFET geometry scaling without explicit physical constraints. These results establish OOD-rewarded quantum architecture search as a framework to discover parameter-efficient surrogate for data-scarce device optimization.

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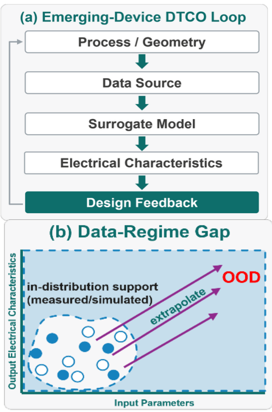
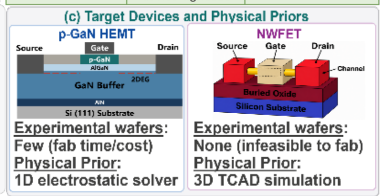


Table 1. Comparisons of different methods

Method	Pros	Cons
Experimental	real device physics	fabrication cost, time-consuming, noisy
TCAD	physics-based, accurate	slow, limited coverage, real-world calibration gap
Conventional ML	fast inference	noise-sensitive, overfits, poor OOD
Proposed Method	noise-robust, strong OOD, device-agnostic	per-device dataset and training required



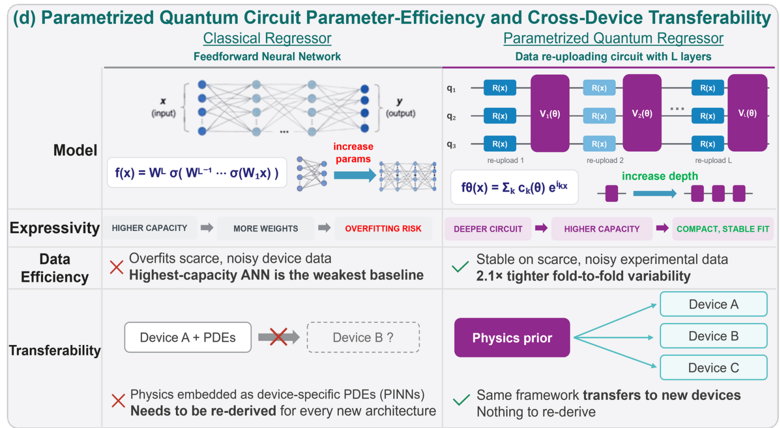


Fig. 1. Emerging-device surrogate-modeling DTCO challenges and the proposed unified OOD modeling framework: (a) conventional DTCO workflow and modeling methods (Table 1); (b) high-value optimization requires physically consistent OOD extrapolation, where CML degrades; (c) target devices and data sources; and (d) data re-uploading PQCs provide parameter-efficient expressivity scaling and enable a common framework that adapts across devices without re-deriving PDEs.

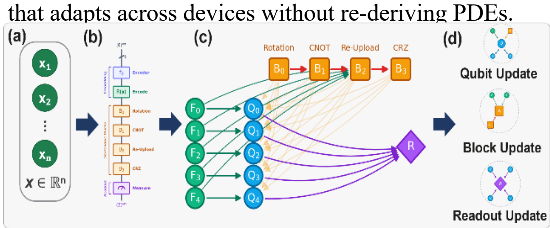


Fig. 2. RL-search framework for PQC architecture discovery. (a) Device features form the input vector. (b) Inputs are encoded into an n-qubit PQC with searchable rotation and entanglement blocks. (c) The PQC is mapped to a heterogeneous graph of feature, qubit, block, and readout nodes. (d) A relational graph-attention network (GNN) predicts the next circuit operation during RL search.

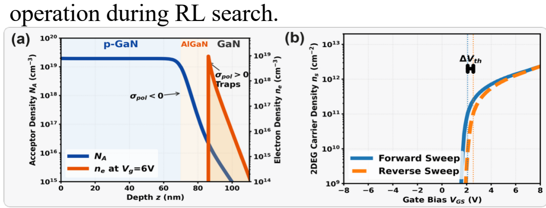


Fig. 3. Electrostatic validation of the p-GaN HEMT solver: (a) doping profile, and (b) simulated 2DEG density versus gate bias.

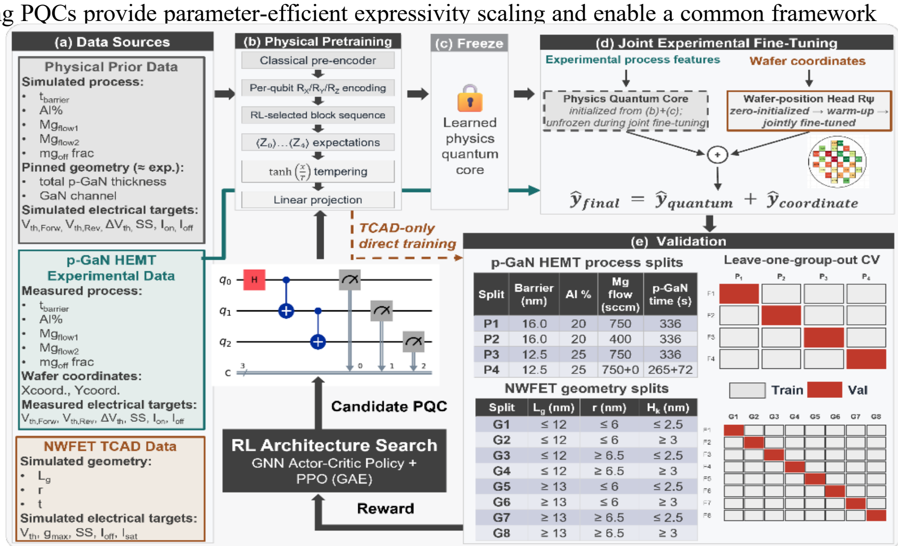


Fig. 4. Training and OOD evaluation workflow for RL-searched quantum surrogate models: (a) physics-prior data for pretraining and experimental data for fine-tuning; (b) simulated-data pretraining initializes the PQC physics core; (c) the pretrained core is frozen; (d) experimental fine-tuning updates the unfrozen core and wafer-position correction head; and (e) nested LOGOCV measures OOD performance and supplies the RL reward. For NWFET, the workflow uses direct TCAD-only training due to lack of available experimental data.

Mg Doping Profile

Continuous Mg Out-Diffusion

$$\sigma(u) = (1 + e^{-u})^{-1} \quad \text{(sigmoid)}$$

$$S_{\text{stage2}} = 1 - \sigma\left(\frac{z - t_{\text{stage2}}}{w_{\text{trans}}}\right)$$

$$S_{\text{stage1}} = \sigma\left(\frac{z - t_{\text{stage2}}}{w_{\text{trans}}}\right) \cdot \left[1 - \sigma\left(\frac{z - t_{\text{pGaN}}}{w_{\text{trans}}}\right)\right]$$

$$S_{\text{tail}} = \sigma\left(\frac{z - t_{\text{pGaN}}}{w_{\text{trans}}}\right) \cdot \exp\left[-\frac{(z - t_{\text{pGaN}})}{\lambda_{\text{tail}}}\right]$$

Acceptor Profile

$$N_A(z) = N_{A,\text{stage2}} \cdot S_{\text{stage2}} + N_{A,\text{stage1}} \cdot \left(S_{\text{stage1}} + f_{\text{tail}} \cdot S_{\text{tail}}\right)$$

$$N_{A,\text{stage2}} \leftarrow \max\left(N_{A,\text{stage2}}, f_{\text{residual}} \cdot N_{A,\text{stage1}}\right)$$

Mg Flow → Acceptor Density

$$N_A(f) = N_{A,0} \cdot f^{\gamma} \ \text{if } f > 0, \text{else } N_{\min}$$

Band Structure and Polarization

Band Edges

$$E_{C0}(z) = -\chi(z)$$

$$E_C(z) = E_{C0}(z) - \varphi(z)$$

$$E_V = E_C - E_g$$

Polarization Sheet Charge

$$P_{\text{sp,AlGaN}} = x \cdot P_{\text{sp,AlN}} + (1 - x) \cdot P_{\text{sp,GaN}}$$

$$Q_{\text{pol}} = P_{\text{sp,GaN}} - \left(P_{\text{sp,AlGaN}} + P_{\text{pz,AlGaN}}(x)\right)$$

$$\sigma_{\text{pol}} = +Q_{\text{pol}} \ \text{at hetero} - \text{interfaces}$$

1D Electrostatics

Poisson Equation

$$\frac{d}{dz}\left(\varepsilon(z) \cdot \frac{d\varphi}{dz}\right) = -\rho(z)$$

Boundary Conditions

$$\varphi_0 = V_G \quad \text{(external bias)}$$

$$\varphi_0 = V_G - V_{bi} - V_{\text{offset}} \quad \text{(flatband)}$$

$$V_{bi} = \chi_{\text{GaN}} + E_{g\text{GaN}} - kT \cdot \ln\left(\frac{p_{\text{surf}}}{N_V}\right) - \Phi_{\text{gate}}$$

$$\varphi_{N-1} = 0 \quad \textit{(backside)}$$

Carrier Statistics $\rho(\varphi)$

Charge Density

$$\rho_i = q\left(p_i - n_i - N_{A,i}^{-}\right) + \frac{\sigma_{\text{pol},i}}{h_i} + \rho_{\text{slow},i}$$

Bulk-Limited Holes

$$p_i = \min\left(N_V \cdot \exp\left(\frac{E_V - E_F}{kT}\right), p_{\text{bulk}}\right)$$

$$p_{\text{bulk}} = \frac{-N_V + \sqrt{N_V^{2} + 4 \cdot g_A \cdot N_A \cdot N_V \cdot \exp\left(\frac{E_A}{kT}\right)}}{2 \cdot g_A \cdot \exp\left(\frac{E_A}{kT}\right)}$$

Ionized Acceptors

$$N_{A,i} = \frac{N_{A,i}}{1 + g_A \cdot \exp\left(\frac{E_{A,\text{lvl}} - E_F}{kT}\right)}$$

2DEG Decay Envelope

$$\psi_{\text{env}}(z) = \frac{\exp\left(\frac{-(z - z_{\text{int}})}{\lambda_{\text{dec}}}\right)}{\int \exp\left(\frac{-(z' - z_{\text{int}})}{\lambda_{\text{dec}}}\right) dz'}$$

$$n_i = N_{2D} \cdot kT \cdot \ln\left(1 + \exp\left(\frac{E_F - E_C}{kT}\right)\right) \cdot \psi_{\text{env}}$$

FV + Newton Solver

Finite-Volume Residual

$$R_i = c_i^R(\varphi_{i+1} - \varphi_i) - c_i^L(\varphi_i - \varphi_{i-1}) + \rho_i$$

$$c_i^L = \frac{\varepsilon_{i-1/2}}{h_i \cdot h_{i,L}}$$

$$c_i^R = \frac{\varepsilon_{i+1/2}}{h_i \cdot h_{i,R}}$$

$$\varepsilon_{i\pm 1/2} = \frac{2 \cdot \varepsilon_i \cdot \varepsilon_{i\pm1}}{\varepsilon_i + \varepsilon_{i\pm1}}$$

Tridiagonal Jacobian

$$J_{i,i-1} = c_i^L$$

$$J_{i,i+1} = c_i^R$$

$$J_{i,i} = -c_i^L - c_i^R + \frac{d\rho_i}{d\varphi_i}$$

Newton Update and Damping

$$J \cdot \delta\varphi = -R$$

$$\delta\varphi \leftarrow \delta\varphi \cdot \min\left(1, \frac{V_{\max}}{\max|\delta\varphi|}\right)$$

$$\text{converge:} \ \max|\delta\varphi| < \varepsilon_{\text{tol}}$$

Newton-Raphson iteration

Fig. 5. Governing equations of the electrostatic solver for p-GaN HEMTs with considering the parameters in Table 2.

Table 2. Parameters used in the solver.

Parameter	Description	Value
$\varepsilon_{r,GaN}$	Relative permittivity of GaN	9.5
$\varepsilon_{r,AlGaN}$	Relative permittivity of AlGaN	9.5−0.5x
$P_{sp,GaN}$	Spontaneous polarization of GaN	−0.034 C/m²
$P_{sp,AlN}$	Spontaneous polarization of AlN	−0.090 C/m²
$P_{pz,AlGaN}$	AlGaN piezoelectric polarization model	$-0.0525x+0.0282x^2$
g_A	Mg acceptor degeneracy factor	4.0
E_A	Mg acceptor activation energy	0.160 eV
N_V	Valence-band effective density of states	2.5×10^{19} cm^{-3}
N_C	Conduction-band effective density of states	2.2×10^{18} cm^{-3}
μ_0	Nominal low-field mobility prefactor	2000 cm²/(V·s)
V_{DS}	Drain bias for low-field current	1.0 V
L_{gate}	Gate length	0.8 μm
$E_{g,GaN}$	GaN energy gap	3.4 eV
χ_{GaN}	GaN electron affinity	4.1 eV
$E_{g,AlN}$	AlN energy gap	6.2 eV
χ_{AlN}	AlN electron affinity	0.6 eV
C_{bow}	AlGaN bandgap bowing parameter	1.0 eV
N_{2D}	2-D density of states in GaN	8.36×10^{13} cm^{-2} eV^{-1}
Φ_{gate}	Schottky gate metal work function	4.8 eV
w_{trans}	Doping transition width	2.0 nm
λ_{tail}	Mg out-diffusion tail length	5.0 nm
f_{tail}	Mg out-diffusion tail fraction	0.02
λ_{decay}	2DEG envelope decay length	3.0 nm
f_{active}	Fixed acceptor activation fraction	0.045
N_t	Slow-trap sheet density (sampled)	10^{11}–10^{13} cm^{-2}
$E_{t,offset}$	Trap energy offset above E_C (sampled)	0.1–0.8 eV
T_s	Trap relaxation time constant (sampled)	10^{-2}–10^{3} s
Mg coeffs	$N_A[\text{cm}^{-3}] = A(\text{Mg flow[sccm]})^B$	$A=3.251\times10^{17}$; $B=0.622$

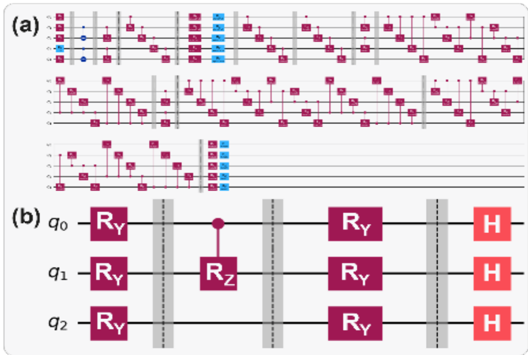


Fig. 6. Representative PQCs discovered by the RL full-fold training for (a) p-GaN HEMTs and (b) NWFETs.

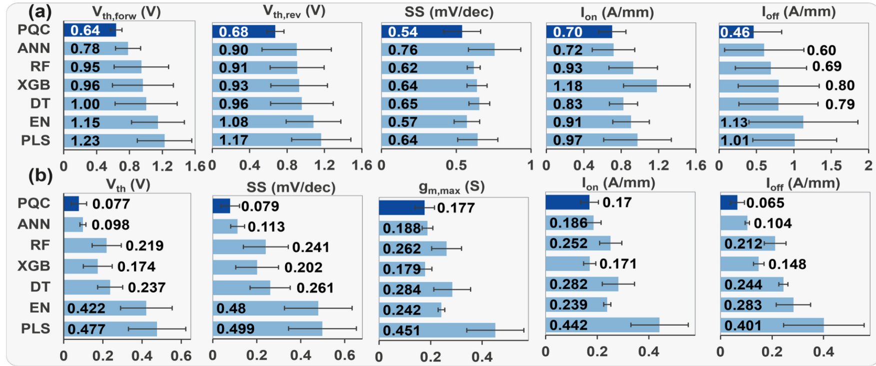


Fig. 7. OOD prediction MAE (z-normalized) comparisons for (a) HEMTs, and (b) NWFETs.

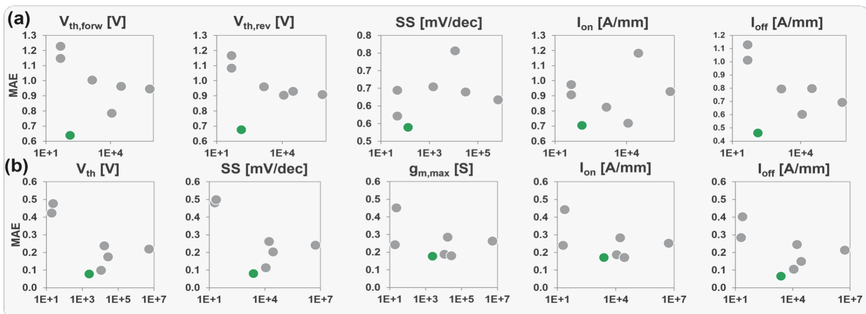


Fig. 8. Pareto-frontier of OOD prediction error z-normalized versus trainable model parameters for (a) p-GaN HEMT and (b) NWFET.

Table 3. GPU speedup for dataset generation/training.

Approach	Hardware	Runtime	Throughput	Speedup
Pretraining Dataset Generation				
Commercial TCAD simulation	CPU	days-months	~10-30 devices/h	**1.0x**
Physics-based solver	CPU	~8.4h	1,785 devices/h	**~90x**
PhysicsNeMo PINN	GPU	~1.2h	12,898 devices/h	**~650x**
Analytical Parameter-shift Model Training				
Standard Library	GPU	~8h	0.076 steps/s	**1.0x**
CUDA-Q	GPU	~0.3h	2.17 steps/s	**~29x**

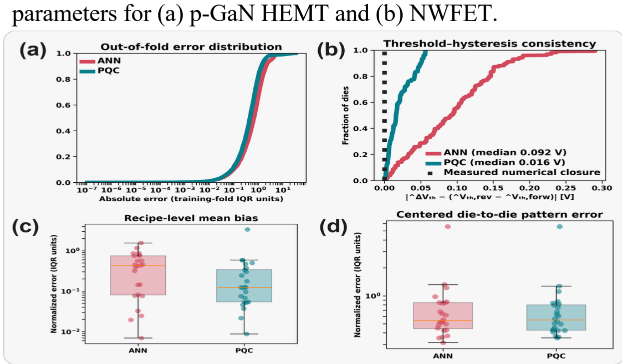


Fig. 9. Comparison of ANN and PQC for p-GaN HEMT OOD prediction: (a) similar out-of-fold error distributions, (b) improved threshold–hysteresis (ΔV_{TH}) consistency, (c) reduced recipe-level bias, and (d) comparable die-to-die variation, indicating PQC gains mainly arise from lower systematic recipe-level bias.

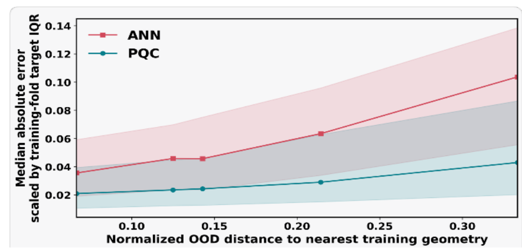


Fig. 10. OOD prediction error versus geometry depth for NWFETs, showing generalization to increasingly distant unseen geometries.

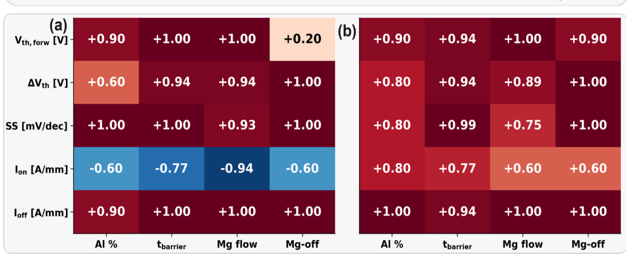


Fig. 11. Retention of solver-learned physical process trends after fine-tuning for (a) ANN and (b) PQC.

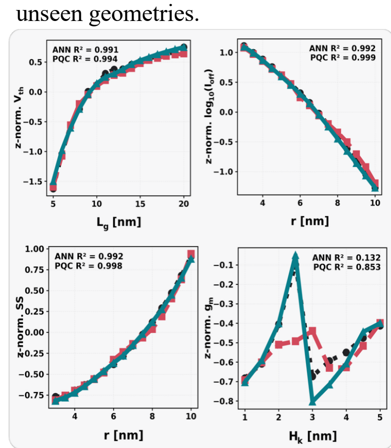


Fig. 12. Geometry-response preservation for held-out NWFET combination, demonstrating higher directional agreement

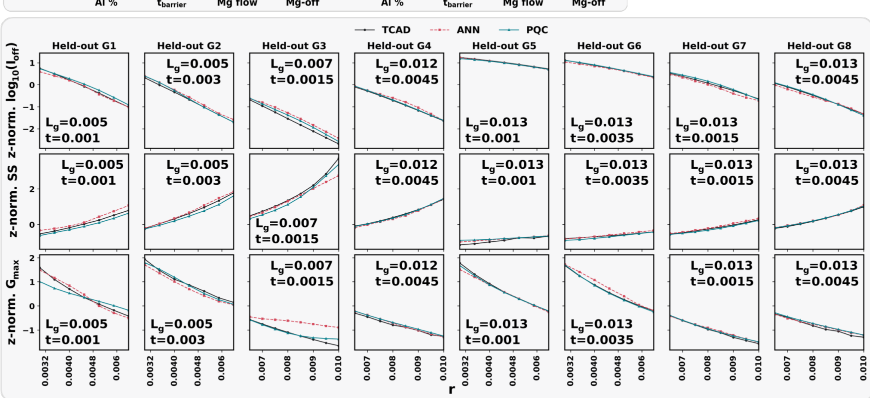


Fig. 13. NWFET out-of-fold predictions (normalized) vs. nanowire radius (r) across characteristics.

Table 4. Comparison with related works

Ref	Dataset	Multi Target	OOD	Quantum component	Physics integration	Scalability
[16]	MTJ experimental + physical model	Yes	No	No	Physical equations embedded in GNN/DNN	Within MTJ/MRAM family
[17]	FeFET TCAD + experimental	Yes	No	No	FeFET physics features embedded in GNN	Within FeFET family; UTBSOI with fine-tuning
[18]	SiC/GaN TCAD + SPICE simulations	Yes	Yes	No	Electro-thermal PDEs embedded in PIGNN	Across WBG families; SiC and GaN devices
[19]	Si NWFET TCAD	Yes	Yes	No	Data-trained physics-informed module	No
[20]	GaN HEMT experimental	No	No	Quantum kernel	No	No
Ours	**p-GaN HEMT experimental + physical model; NWFET TCAD**	**Yes**	**Yes**	**RL-searched VQC**	**No device-specific equations embedded**	**Across distinct device families**